

113B Data Science and Python Programming

Homework Assignment #1

Due: 2025/3/26 12:00:00

Since this is an advanced course for the student who has already passed the introductory course of basic programming with Python, in this homework, you are challenged to write a program that returns the answer to the corresponding question.

Problem #1: Stochastic Program for a Discrete Random Variable (50%)

Let X be a discrete random variable with the following probability mass function (PMF):

$$P_X(k) = \begin{cases} \frac{1}{7}, & k = 4 \\ \frac{1}{4}, & k = 7 \\ \frac{2}{7}, & k = 8 \\ \frac{9}{28}, & k = 9 \\ 0, & \text{otherwise} \end{cases}.$$

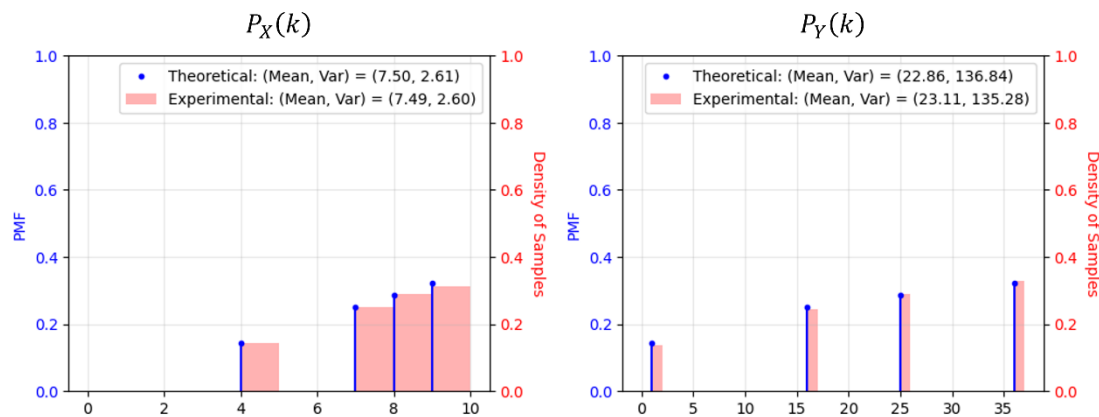
Please write a program that meets the following requests:

1. The program is capable to produce N random samples from $P_X(k)$.
2. Plot a figure that contains the $P_X(k)$ and the histogram of these N random samples.
3. Calculate and return the **mean and variance** of $P_X(k)$ for **both theoretical and experimental values**. Either embedding this message in your figure or in terminal is acceptable.
4. If there is a new random variable $Y = (X - 3)^2$, repeat the above 3 requests for $P_Y(k)$.

Here is the sample code:

```
class solution1:
    def __init__(self, numSamples=1000, k=[4, 7, 8, 9], p=[1/7, 1/4,
2/7, 9/28]):
        self.N = numSamples
        self.k = np.array(k)
        self.p = np.array(p)
        剩下你自己想喔
```

If everything goes well, you might have these result:



FRIENDLY REMINDER

You don't need to follow the figure style.

Feel free to create your own style!

Problem #2: The Central Limit Theorem (50%)

Let $X_1, X_2, \dots, X_n$ be a continuous random sample from the following probability density function (PDF):

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & 0 \leq x \leq 1. \\ 0, & \text{otherwise} \end{cases}$$

Please write a program that meets the following requests:

1. The program is capable to produce N random samples from $f_X(x)$. You can produce the random sample X by $\sqrt{2U + \frac{1}{4}} - \frac{1}{2}$, where $U \sim \text{Uniform}(0,1)$.
2. Plot a figure that contains the $f_X(x)$ and the histogram of these N random samples.
3. Calculate and return the **mean and variance** of $f_X(x)$ for **both theoretical and experimental values**. Either embedding this message in your figure or in terminal is acceptable.
4. Let Y be a new random variable defined as followed,

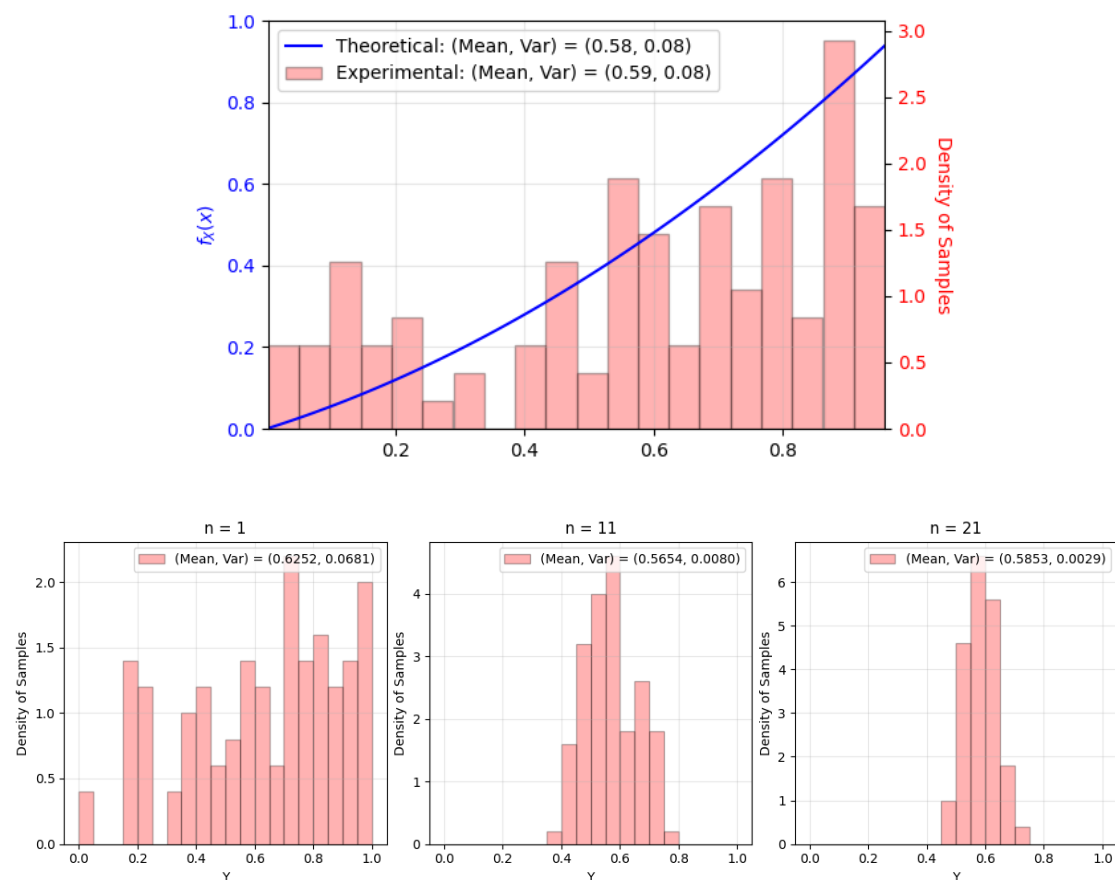
$$Y = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

Plot the histogram of Y with N random samples and calculate its mean and variance under different values of n . Hint: for the case of $n = 2$, you first generate N random samples from $f_X(x)$ as X_1 . Then generate another N random samples from $f_X(x)$ as X_2 to get N random samples of $Y = \frac{X_1 + X_2}{2}$.

Here is the sample code:

```
class solution2:
    def __init__(self, numSamples=1000):
        self.N = numSamples
        self.samples = self.pdf()
        剩下你自己想喔
```

If everything goes well, you might have these result:



Again, you don't need to follow the figure style. Feel free to create your own style. More importantly, what do you observed in this figure?

Please accomplish this homework with an organized code (e.g., with main script and function script). Here is a template for your code structure:

```
113B_hw1_0123456789
├─ ans.py          # Solution of homework #1
├─ xxx.py          # xxx module if necessary
└─ main_hw1.py     # Main script of homework #1
```

You don't need to follow this structure, just keep your main script clean.

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs (e.g., signature). **Your signature should be attached in your main script.** Here is the template of signature:

```
"""
Created on Wed Aug 7 01:23:45 2024

@author: Xi Winnie, student ID

@collaborators: William Lai, his student ID
"""
```

Please save your code as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113B_hw1_**ID**.zip

For example,

113B_hw1_**0123456789**.zip

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “E3@NYCU” system. There will be a section for uploading your homework.